

Xiao Yan

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Location: Dallas, TX

Personal Website

Department of Computer Science, UT Dallas

Education

University of Texas at Dallas

Ph.D. in Computer Science, Systems Track — On-Device Intelligence, Edge-Cloud Systems

08/2024 – Present

Qualifying exam passed April 2026

Imperial College London

M.Sc. Applied Computational Science & Engineering with Merit

10/2022 – 10/2023

University of Birmingham

B.Sc. Computer Science with Honours, First Class

08/2019 – 07/2022

Publications

Published

- **X. Yan** and Y. Ding, “Are We There Yet? A Measurement Study of Efficiency for LLM Applications on Mobile Devices,” in *Proc. 2nd International Workshop on Foundation Models for Cyber-Physical Systems & IoT (FMSys '25)*, 2025.

Under Review

- **X. Yan**, C. Htun, J. Yang, Z. Liu, D. Li, and Y. Ding, “Hestia: Semantic Acoustic Sensing and On-Device LLM Reasoning for Privacy-Preserving Home Routine Recognition” (under review, ACM SenSys).

Research Experience

Multimodal Edge Sensing for Wildlife Monitoring (Field Deployment)

06/2026 – Present

The University of Texas at Dallas

- Leading the multimodal systems track of a wildlife-monitoring field deployment combining camera, acoustic, PIR, and satellite-RF (GNSS) sensing on autonomous solar-powered edge nodes.
- Implemented a two-tier architecture that moves computation toward the sensing front-end: sub-watt ESP32 nodes wake a Raspberry Pi 5 for MobileNetV2 inference, projecting a several-fold energy reduction versus always-on capture.
- Built and bench-validated the full pipeline end-to-end, using instrumented power profiling of each sensor and compute stage to guide the hardware/software co-design.

Privacy-Preserving Home Routine Recognition via Acoustic Sensing

08/2025 – 02/2026

The University of Texas at Dallas

- Designed Hestia, a hybrid edge-central system that reconstructs daily routines from sparse acoustic detections using commodity microphones and on-device LLM reasoning, with no raw audio leaving the home.
- Developed hierarchical temporal clustering, a deterministic scaffold that caps on-device LLM calls per day, and continual self-adaptation via LLM pseudo-labels.
- Deployed and evaluated across five real homes (17 subject-days): acoustic sensing tracked a co-deployed camera baseline and outperformed WiFi CSI and IMU, while keeping all raw audio in the home.

Efficiency of LLM Applications on Mobile Devices

01/2025 – 03/2025

The University of Texas at Dallas

- Measured efficiency tradeoffs across mobile, edge, and cloud deployments for LLM-based applications, benchmarking end-to-end latency and memory across model families, sizes, and compression strategies.
- Built AutoLife-Lite, a lightweight LLM application that infers user location and activity contexts from smartphone sensor data in real time.

Emotion-Aware Chatbot with Facial Expression Recognition

09/2021 – 03/2022

University of Birmingham

- Built a multimodal chatbot fusing real-time CNN facial expression recognition with LSTM dialogue modeling; deployed end-to-end as a standalone application.

Technical Skills

Programming Languages: Python, Java, C/C++, MATLAB, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn

On-Device Inference: ExecuTorch, MediaPipe LiteRT, TFLite, Ollama / llama.cpp, MNN; 4-bit quantization

Tools & Platforms: Git, Linux, LaTeX, Docker, Android Studio, Raspberry Pi, ESP32

Domains: Mobile Computing, On-Device LLM Systems, Edge-Cloud Architecture, Acoustic Sensing

Languages: English (Professional), Mandarin Chinese (Native)